

Marketplace | Customer Satisfaction Analysis | Olist Dataset (2016-2018)

Analysis of 93,000+ delivered orders on the hypothetical *Cartana* B2C marketplace reveals that customer satisfaction outcomes are not evenly distributed. The platform average of 4.15 conceals meaningful variation, primarily explained by seller-level performance differences. **57.3% of delivered orders flow through the weaker half of sellers, and buyers exposed to those sellers experience double the bad review rate (16.2% vs 7.7%).** This is a systemic exposure problem, not an edge case.

Business Context

Cartana's VP of Product commissioned this analysis after CSAT scores declined for two consecutive quarters. In this case study, the Growth Analytics team was tasked with identifying primary drivers of poor customer experience and recommending targeted interventions.

Three guiding questions structured the investigation:

- **What drives satisfaction in customer satisfaction?**
- **How does delivery performance impact CSAT?**
- **Is seller performance a key mechanism behind issues?**

Data and Methodology

Dataset: Olist Brazilian E-Commerce Public Dataset (Kaggle, CC BY-NC-SA 4.0). Analysis in Python using **pandas** and **Plotly**.

- 93,412 delivered orders retained after filtering cancelled/undelivered records and delivery delta outliers (97.47% retained).
- Review deduplication: most recent review kept per order_id (1,098 duplicate rows resolved across 547 orders).
- Delivery delta = actual delivery date minus estimated. Positive = late.
- Seller tiers defined by median split on seller-level average review score (with a 4.23 threshold). Min. 10 orders per seller.

Key Findings

Finding	Headline	Detail
Finding 1	Satisfaction varies significantly across the marketplace	Overall avg 4.15, NPS proxy 66.1. Category scores range from 3.66 (office_furniture) to 4.55 (books_general_interest) — 0.89-point spread. Physical and complex product categories consistently underperform.

Finding 2	Delivery performance is the primary operational driver	On-time avg 4.29 stars → 7+ days late avg 1.70 stars: a 2.59-point drop. Late orders are 4.9x more common among bad reviews. Late volume is 6.8% of orders but 33% of bad reviews.
Finding 3	Seller performance variation is the mechanism	Low-performing sellers generate 2x the bad review rate (16.2% vs 7.7%) and 68% higher late delivery rates (8.0% vs 4.8%). They account for 57.3% of delivered orders — systemic, not marginal.

Recommendations

Recommendation	Action	Expected Impact	Effort
01 Seller Quality Visibility for Buyers	Visible seller reliability indicator (review score + on-time rate) at point of purchase.	Reduced buyer exposure to low-performing sellers	Medium
02 Proactive Communication for At-Risk Deliveries	Flag high-risk orders and notify buyers proactively with a revised delivery estimate.	Reduction in 1-star reviews from unmet expectations	Medium
03 Seller Performance Feedback Loop	Monthly score summary to bottom-quartile sellers vs category avg. Pair with outreach for sellers below 3.5.	Improved seller awareness with downstream effect on satisfaction	Low-Med

Limitations and Next Steps

- Dataset covers 2016-2018 only. This is considered a historical sample.
- Seller tier defined by median split, which is a simplification.
- Repeat purchase signal is weak in Olist by design, so that has been disregarded.
- The recommendations consider hypothetical flaws. A/B testing the seller quality indicator on a category subset would validate the mechanism.